MAHMOUD KHAIRY ABDELGHANY AMER

Automation & AI QA Specialist | Senior QA & Automation Expert | Quality Automation & AI Lead

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PROFILE SUMMARY

Highly accomplished Automation and AI-Driven Testing Professional with 8+ years delivering scalable QA solutions across 20+ projects valued at $18B+. Enhanced test efficiency by 40%, accelerated defect detection by 35%, and improved regression coverage by 30% through AI-powered platforms, predictive analytics, and automation frameworks. Expert in building cross-functional teams, implementing intelligent testing strategies, and leveraging tools such as Selenium, Robot Framework, Playwright, Appium, Power BI, and GPT-powered solutions. Recognized for driving innovation, process optimization, and knowledge-sharing programs that consistently improve quality, reduce errors, and achieve measurable business impact.

CORE QUALIFICATIONS

- Expert in designing and implementing AI-driven automation frameworks to improve testing accuracy and scalability across enterprise applications.
- Skilled in developing intelligent test agents, self-healing scripts, and predictive analytics models to optimize QA workflows.
- Proficient in end-to-end test management, including API, performance, and cross-platform testing for web, mobile, and desktop environments.
- Strong experience in CI/CD and DevOps integration, leveraging tools like Azure Pipelines, Docker, Git, and YAML for seamless deployment and automation.
- Adept at mentoring and leading cross-functional teams, fostering knowledge-sharing, continuous learning, and adoption of innovative QA practices.
- Competent in creating interactive dashboards, reporting metrics, and data-driven insights to support strategic decision-making and process improvements.

SKILLS

- Automation Frameworks
- AI Integration
- Test Optimization
- Predictive Analytics
- Regression Testing
- API Validation
- Performance Testing
- Cross-Platform
- CI/CD Implementation
- Data Analytics
- Defect Management
- Dashboard Reporting
- Process Improvement
- Team Mentorship
- Test Automation
- Intelligent Agents
- Risk Assessment
- Knowledge Sharing

PROFESSIONAL EXPERIENCE

Staff Automation Engineer **Jan 2024 – Present**

Thiqah Business Services, Riyadh, Saudi Arabia

Lead AI-driven automation initiatives, enhancing testing efficiency, team engagement, and organizational AI adoption.

Key Achievements

- warded Employee of the Year (2024) within the Quality Team for technical leadership, innovation in AI automation, and elevating team performance.
- Built and led the development of “Quality Mate”, an AI-powered automation platform integrated with Azure DevOps, Flowise, and GPT, delivering end-to-end traceability, automated test case generation, bug logging, and script creation, improving testing efficiency by 40%.
- Implemented AI-driven practices including self-healing scripts, AI locators, predictive test selection, and intelligent dashboards, accelerating defect detection by 35% and improving regression coverage by 30%.
- Launched gamification programs such as Test League and QC Tube, increasing cross-team engagement and knowledge sharing by 50%.
- Coordinated and mentored Thiqah Hackathon (Nov 2025), guiding 10+ cross-functional teams on AI implementations, resulting in high-quality prototype delivery.
- Secured 2nd Place - Elm Hackathon (Dec 2025) with team “Agent X” by developing an AI-driven solution, demonstrating rapid prototyping and problem-solving skills under tight deadlines.

Lead Automation Testing Engineer **Sep 2022 – Jan 2024**

Thiqah Business Services, Riyadh, Saudi Arabia

Designed and optimized automation frameworks to enhance QA efficiency and team productivity.

Key Achievements

- Developed and optimized Robot Framework, Playwright, Cypress, and Appium frameworks, reducing test cycle time by 30% across 10+ projects.
- Executed 2,000+ automated test cases, achieving 95% functional coverage and increasing defect detection by 25%.
- Mentored a team of 5 engineers, boosting skill competency and collaboration, raising overall productivity by 20%.

Senior Quality Control Engineer **Apr 2019 – Sep 2022**

Thiqah Business Services, Riyadh, Saudi Arabia

Delivered quality assurance for high-impact government projects, ensuring compliance, performance, and reliability.

Key Achievements

- Designed detailed test cases and executed performance tests for 15+ government projects, reducing post-release defects by 30%.
- Implemented Agile QA practices, improving sprint delivery predictability by 20% and accelerating communication between development and QA teams.
- Managed TFS and MTM test cases, increasing traceability of critical defects by 40%.

Senior Quality Control Engineer **Mar 2018 – Apr 2019**

Algorithmz, Riyadh, Saudi Arabia

Performed comprehensive software testing across multiple platforms, ensuring seamless user experience and system reliability.

Key Achievements

- Conducted cross-platform testing on web, Android, iOS, and Smart TV applications, improving functional reliability by 25%.
- Developed detailed bug reports for 200+ defects, accelerating resolution time by 30%.
- Executed API and performance testing, reducing post-deployment issues by 20%.

Quality Control Engineer II **Feb 2017 – Mar 2018**

Path Solutions, Riyadh, Saudi Arabia

Executed functional and regression testing for financial applications, ensuring system accuracy and compliance.

Key Achievements

- Developed and executed 1,500+ manual and automated test scripts, validating backend and frontend financial systems.
- Performed SQL-based database testing, improving data integrity accuracy by 30%.
- Collaborated across 5+ projects simultaneously, achieving 100% on-time delivery and reducing production defects by 15%.

Quality Control Engineer I **May 2016 – Feb 2017**

Silicon Expert Technologies (Arrow), Riyadh, Saudi Arabia

Validated database tools and ensured data quality across large-scale analytics and operational processes.

Key Achievements

- Reviewed and validated 500+ new database entries, improving workflow accuracy by 20%.
- Designed and executed testing procedures, reducing tool defects by 25%.
- Conducted statistical analysis and generated executive reports, supporting data-driven decisions and improving operational efficiency by 15%.

EDUCATION

Bachelor of Science - Electronic Engineering, Communication Department **2014**

Menoufia University, Egypt

- Graduation Project: GSM Based Home Automation using Arduino programmed in C language - Excellent

PROJECTS & KEY INITIATIVES

AI Quality Mate Platform - Creator & Lead Architect

- Architected and delivered an AI-driven automation platform integrating Azure DevOps, Flowise, and GPT models to streamline QA workflows.
- Implemented end-to-end test lifecycle automation including test case generation, task assignment, defect logging, script creation, and reporting with full traceability.
- Significantly improved testing efficiency, reduced manual workload, and positioned the organization at the forefront of AI-enabled QA innovation.

AI-Powered Testing Agents - Innovation Lead

- Designed and deployed intelligent testing agents featuring self-healing scripts, predictive analytics, and real-user simulation capabilities.
- Increased test stability, accelerated defect detection cycles, and optimized regression execution performance.

Test League - QA Gamification Initiative

- Launched an interactive gamified learning ecosystem featuring quizzes, challenges, and performance leaderboards for QA engineers.
- Strengthened collaboration, engagement, and cross-team knowledge sharing across manual and automation testing groups.

QC Tube Knowledge Hub - Founder

- Established an internal learning platform delivering workshops, webinars, and technical sessions on testing, automation frameworks, and AI solutions.
- Fostered a culture of continuous learning, technical excellence, and innovation across engineering teams.

Custom GPT Ecosystem for QA Support - Developer

- Built and deployed a suite of five specialized GPT tools to enhance productivity and technical growth for QC engineers.
- Solutions included: ISTQB Study Partner, QC Senior Interviewer, User Story & Test Case Generator, SQL Query Generator, and Python Automation Assistant.
- Reduced routine coding effort and enabled real-time technical guidance for engineers at all levels.

Automation Dashboards & Metrics - Data Strategy Initiative

- Developed interactive analytics dashboards using Power BI, Metabase, and custom tools to monitor automation coverage, defect rates, execution speed, and ROI.
- Enabled leadership to make real-time, data-driven quality and delivery decisions.

Hackathons & Innovation Programs - Contributor & Mentor

- 2nd Place Winner - Elm Hackathon (Dec 2025): Built an AI-driven solution with Team Agent X, demonstrating rapid prototyping and advanced problem-solving under tight deadlines.

- Organizer & Mentor - Thiqah Hackathon (Nov 2025): Coordinated event execution and mentored cross-functional teams in AI solution design.
- Contributed to multiple innovation initiatives involving AI-powered locators, predictive testing systems, and chatbot integrations to drive organizational innovation.
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CERTIFICATIONS & PROFESSIONAL TRAINING

ISTQB & Professional Certifications

- **ISTQB® Advanced Level - Test Manager** — **Jan 2025**
- **ISTQB® Foundation Level** — **May 2018**
- **Strategic Planning & Performance Management** — **May 2024**

AI / ML & Emerging Technologies

- **Full Stack AI Course** — **May 2023**
- **ChatGPT Impact on STLC** — **Jan 2023**
- **Automation Testing using AI/ML Technology** — **Oct 2022**
- **AI/ML Impact on Software Testing Life Cycle** — **May 2022**

Automation Frameworks & Tools

- **Automation with Playwright Framework** — **Jun 2023**
- **Integration Appium with Robot Framework** — **Jun 2021**
- **Automation Testing using Robot Framework & Python** — **Jul 2019**
- **Mobile Automation using Appium** — **Sep 2018**
- **Web Automation using Selenium WebDriver** — **Aug 2018**
- **Fully Automated API Testing Cycle** — **Feb 2022**

Testing Tools & Technologies

- **Performance Testing using JMeter** — **Mar 2022**
- **API Testing with Postman** — **Oct 2018**
- **XML & JSON** — **Oct 2018**
- **Database Testing** — **Jun 2018**

Programming & Technical Foundations

- **SQL** — **Jan 2018**
- **Java SE** — **Dec 2017**
- **HTML** — **Jun 2018**
- **ICDL** — **Jan 2012**

TECHNICAL SKILLS

Automation Tools & Frameworks

- Selenium WebDriver, Robot Framework, Playwright, Cypress, Appium, Katalon Studio, Testim, Sofy.ai, UiAutomatorViewer, BrowserStack, Zalenium

AI ML & Intelligent Testing

- Quality Mate creator & lead, ChatGPT, Flowise, Applitools, AI-driven locators, self-healing scripts, predictive analytics agents, AI-powered dashboards

API Testing

- Postman, SoapUI, Swagger

Performance Testing

- JMeter, LoadRunner

Databases & Data Testing

- SQL Server, MySQL, MongoDB, Toad

CI CD & DevOps

- Azure Pipelines, Git, Docker, Linux, YAML, PuTTY

Programming & Scripting

- Python, Java SE, SQL, HTML, JSON, XML

Test Management & Bug Tracking

- Azure DevOps TFS MTM, JIRA

Analytics & Reporting

- Power BI, Metabase, Custom dashboards for test metrics and ROI tracking

Other Tools & Platforms

- Microsoft Office Word Excel PowerPoint, PyCharm IDE

LANGUAGES

Arabic- Native | **English** - B2 (Intermediate)